Summations

Part VIII Appendix

Scope

Derive arithmetic, power, geometric, telescoping, and product identities from finite sums. Analyze loops with fixed-constant induction, tail ratios, dyadic blocks, and integral bounds. State convergence assumptions before using infinite series or changing summation order.

Prerequisites

Arithmetic and geometric sequences, logarithms, and integral estimates.

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Appendix route

Learning objectives

- Use summation formulas and linearity.
- Bound sums by integration, splitting, and geometric series.
- Distinguish finite sums from infinite series.
- A.1 Summation formulas and properties
- A.2 Bounding summations

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Summation formulas and properties

The finite sum $\sum_{k=1}^{n} a_k$ equals 0 when $n = 0$. The finite product $\prod_{k=1}^{n} a_k$ equals 1 when $n = 0$. These empty expressions use the additive and multiplicative identities, so formulas also cover loops with zero iterations. Finite terms may be reordered arbitrarily, and

$$\sum_{k=1}^{n}(ca_k + b_k) = c\sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k.$$

An infinite series $\sum_{k=1}^{\infty} a_k$ is the limit of its partial sums S_n. It is not obtained by substituting ∞ into a finite formula. If the partial sums have no finite limit, the series diverges. Convergence requires $a_k \to 0$, because $a_k = S_k - S_{k-1}$. That condition is insufficient. The terms $1/k$ tend to zero, but the harmonic series diverges.

Absolute convergence means that $\sum_{k=1}^{\infty} |a_k|$ converges. It permits arbitrary reordering without changing the sum. A conditionally convergent series, such as $\sum_{k=1}^{\infty}(-1)^{k+1}/k$, need not preserve its sum under reordering, which changes partial-sum cancellation. Termwise calculus below requires a convergence interval. Finite linearity does not justify arbitrary infinite operations.

Arithmetic sums by pairing and induction

Write $S_n = 1 + 2 + \cdots + n$ and reverse the order to obtain $n + (n-1) + \cdots + 1$. Adding the two expressions gives $2S_n = n(n+1)$, so

$$S_n = \frac{n(n+1)}{2}.$$

The formula includes $n = 0$. Induction gives a second verification. The base case is $S_0 = 0$, and the induction step is

$$S_{n+1} = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

The induction coefficient stays fixed as n changes. More generally,

$$\sum_{j=0}^{n-1}(a + jd) = na + \frac{dn(n-1)}{2}.$$

A loop with $j = 3, \ldots, 7$ that performs $2j - 1$ units of work has total cost

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Arithmetic sums by pairing and induction (continued)

$$\begin{aligned}\sum_{j=3}^{7}(2j-1) &= \left(2\sum_{j=1}^{7} j - 7\right) - \left(2\sum_{j=1}^{2} j - 2\right) \\ &= 49 - 4 = 45 \\ &= 5 + 7 + 9 + 11 + 13.\end{aligned}$$

Changing the starting index requires subtracting the correct prefix. The upper index 7 is not the number of terms, which is 5.

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Square and cube sums from telescoping

Sum $(k+1)^3 - k^3 = 3k^2 + 3k + 1$ for $k = 0, \ldots, n$. The left side telescopes, giving

$$(n+1)^3 = 3\sum_{k=0}^{n} k^2 + \frac{3n(n+1)}{2} + (n+1).$$

Solving for the sum gives

$$\sum_{k=0}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Similarly, sum $(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1$ and substitute the preceding square and arithmetic sums:

$$(n+1)^4 = 4\sum_{k=0}^{n} k^3 + 6\sum_{k=0}^{n} k^2 + 4\sum_{k=0}^{n} k + (n+1),$$

$$\sum_{k=0}^{n} k^3 = \frac{n^2(n+1)^2}{4} = \left(\frac{n(n+1)}{2}\right)^2.$$

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Square and cube sums from telescoping (continued)

Both formulas hold at $n = 0$. At $n = 4$, the checks are

$$0 + 1 + 4 + 9 + 16 = 30 = \frac{4 \cdot 5 \cdot 9}{6},$$

$$0 + 1 + 8 + 27 + 64 = 100 = \left(\frac{4 \cdot 5}{2}\right)^2.$$

Telescoping proves the formulas for general n. The small cases check arithmetic and transcription.

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Finite, infinite, and boundary geometric sums

Let $G_n(x) = \sum_{k=0}^{n} x^k$. Subtract G_n from xG_n. Interior terms cancel, so

$$(x-1)G_n(x) = x^{n+1} - 1.$$

For $x \neq 1$, this gives $G_n(x) = (x^{n+1} - 1)/(x-1)$. At $x = 1$, evaluate separately to obtain $G_n(1) = n + 1$; division by zero is invalid.

If $|x| < 1$, then $x^{n+1} \to 0$, and

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}.$$

At $x = 1$, partial sums are unbounded. At $x = -1$, they alternate between 1 and 0. If $|x| > 1$, the terms do not tend to zero.

For $x = -1/2$ and $n = 3$, the finite sum is $1 - 1/2 + 1/4 - 1/8 = 5/8$, whereas the infinite sum is $2/3$. When $|x| < 1$, the exact tail and its absolute bound are

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Finite, infinite, and boundary geometric sums (continued)

$$\sum_{k=n+1}^{\infty} x^k = \frac{x^{n+1}}{1-x}, \qquad \left|\frac{x^{n+1}}{1-x}\right| \leq \frac{|x|^{n+1}}{1-|x|}.$$

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Weighted geometric sums and valid calculus

Finite polynomials may be differentiated term by term. A direct derivation starts with $W_n = \sum_{k=0}^{n} kx^k$. Subtracting xW_n gives

$$(1-x)W_n = x + x^2 + \cdots + x^n - nx^{n+1}.$$

Substitute the finite geometric formula and rearrange:

$$W_n = \frac{x - (n+1)x^{n+1} + nx^{n+2}}{(1-x)^2}, \qquad x \neq 1.$$

At $x = 1$, $W_n = n(n+1)/2$. For $|x| < 1$, $n|x|^n \to 0$, because an exponential with a fixed base dominates a polynomial. Taking the limit gives

$$\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2}.$$

This derivation does not interchange differentiation with an infinite sum without proof. Power series permit termwise differentiation and integration inside their radius of convergence, which is 1 here. Endpoints require separate checks.

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Weighted geometric sums and valid calculus (continued)

Apply $x\,d/dx$ once more inside that radius:

$$\sum_{k=0}^{\infty} k^2 x^k = \frac{x(1+x)}{(1-x)^3}.$$

Thus $x = 1/2$ gives $\sum_{k=0}^{\infty} k/2^k = 2$ and $\sum_{k=0}^{\infty} k^2/2^k = 6$. Terms tending to zero alone would not establish these convergent sums.

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Telescoping leaves boundary terms

Each interior term occurs once with each sign, so

$$\sum_{k=1}^{n} (a_k - a_{k-1}) = a_n - a_0.$$

Align indices and endpoints before cancellation. For example,

$$\frac{1}{k(k+2)} = \frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+2}\right),$$

which gives

$$\sum_{k=1}^{n} \frac{1}{k(k+2)} = \frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right).$$

At $n = 3$, $1/3 + 1/8 + 1/15 = 21/40$, matching $\tfrac{1}{2}(3/2 - 1/4 - 1/5)$. Letting $n \to \infty$ gives $3/4$, since the terminal denominators tend to infinity.

Products telescope as well. For $n \geq 2$,

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Telescoping leaves boundary terms (continued)

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)=\left(\prod_{k=2}^{n}\frac{k-1}{k}\right)\left(\prod_{k=2}^{n}\frac{k+1}{k}\right)$$
$$=\frac{1}{n}\frac{n+1}{2}=\frac{n+1}{2n}.$$

At $n=1$, the empty product is 1, as the formula also predicts.

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Products as sums and indispensable premises

If every $a_k > 0$, then

$$\ln\left(\prod_{k=1}^{n} a_k\right) = \sum_{k=1}^{n} \ln a_k.$$

A zero or negative factor has no real logarithm. For example,

$$\prod_{k=1}^{n} 2^{3k+1} = 2^{\sum_{k=1}^{n}(3k+1)} = 2^{3n(n+1)/2+n}.$$

For $0 \le u_k < 1$, logarithm bounds give

$$-\frac{u_k}{1-u_k} \le \ln(1-u_k) \le -u_k.$$

These bounds control products through sums. As u_k approaches 1, the lower-bound denominator cannot be treated as a fixed constant.

If fixed constants satisfy $0 < c_1 f(k) \le h(k) \le c_2 f(k)$ for every $k \ge 1$, summing gives

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Products as sums and indispensable premises (continued)

$$c_1 \sum_{k=1}^{n} f(k) \leq \sum_{k=1}^{n} h(k) \leq c_2 \sum_{k=1}^{n} f(k).$$

If the inequalities hold only on a tail, retain the finite prefix separately. When $\sum_{k=1}^{n} f(k) \to \infty$ and the prefix is finite, the prefix does not change the Θ order. Constants must be uniform over k. An unbounded choice of a different constant at every index proves no common Θ bound.

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Bounding summations

Decide whether an exact value, an upper bound, or a two-sided order is needed. Induction, termwise bounds, fixed ratios, splitting, and integration serve different purposes. A rough upper bound alone does not establish tightness.

For $S_n = \sum_{k=0}^{n} 4^k$, seek $S_n \le c4^n$. The base case $n = 0$ requires $c \ge 1$. The induction step gives

$$S_{n+1} \le c4^n + 4^{n+1} = \left(\frac{c}{4} + 1\right) 4^{n+1}.$$

It closes when $c/4 + 1 \le c$, or $c \ge 4/3$. The fixed choice $c = 4/3$ works for all n. The last term gives $S_n \ge 4^n$, so $S_n = \Theta(4^n)$.

The false induction claiming that $\sum_{k=1}^{n} k = O(n)$ remains linear after adding $n + 1$ hides a growing constant. An assumption $\sum_{k=1}^{n} k \le cn$ would require

$$cn + n + 1 \le c(n + 1), \qquad \text{or} \qquad n + 1 \le c.$$

No fixed c satisfies this for every n.

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Maximum-term upper bounds and tail lower bounds

For nonnegative a_k and any subset $J \subseteq \{1, \dots, n\}$,

$$\sum_{k=1}^{n} a_k \le n \max_{1 \le k \le n} a_k, \qquad \sum_{k=1}^{n} a_k \ge \sum_{k \in J} a_k.$$

Upper bounds may use all terms. Lower bounds often retain enough large terms. For example, $\sum_{k=1}^{n} k \le n^2$. The indices $\lceil n/2 \rceil, \dots, n$ contain at least $n/2$ terms, each at least $n/2$. Thus $\sum_{k=1}^{n} k \ge n^2/4$ for both odd and even n, proving $\Theta(n^2)$ without an exact formula.

For a fixed real $r \ge 0$, the same argument gives

$$2^{-r-1} n^{r+1} \le \sum_{k=1}^{n} k^r \le n^{r+1}.$$

Hence the sum is $\Theta(n^{r+1})$. Fixing r is essential. If $r = n$, the lower coefficient 2^{-r-1} depends on n, and this proof gives no uniform order.

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A geometric tail requires a fixed ratio

Suppose $a_k \geq 0$ and a fixed $0 < \rho < 1$ satisfies $a_{k+1} \leq \rho a_k$ for every $k \geq K$. Induction gives $a_{K+j} \leq a_K \rho^j$, so

$$\sum_{k=K}^{\infty} a_k \leq \frac{a_K}{1-\rho}.$$

The multiplicative inequality also covers zero terms without division by zero.

For $a_k = k^2/3^k$ and $k \geq 2$,

$$\frac{a_{k+1}}{a_k} = \frac{1}{3}\left(1+\frac{1}{k}\right)^2 \leq \frac{3}{4}.$$

The prefix at $k = 0, 1$ contributes $1/3$. The tail starts at $a_2 = 4/9$, giving

$$\sum_{k=2}^{\infty} a_k \leq \frac{4/9}{1/4} = \frac{16}{9}, \qquad \sum_{k=0}^{\infty} a_k \leq \frac{19}{9}.$$

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A geometric tail requires a fixed ratio (continued)

The exact weighted geometric formula gives $3/2$, a sharper finite bound. Harmonic ratios $k/(k+1) < 1$ tend to 1, so no fixed $\rho < 1$ bounds the whole tail. A prefix that depends on n, such as two terms each equal to n^2, is not $O(1)$ merely because it contains only two terms.

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Dyadic blocks reveal harmonic layers

Let $H_n = \sum_{k=1}^{n} 1/k$. The block $k = 2^j, \ldots, 2^{j+1} - 1$ has 2^j terms, each between $1/2^{j+1}$ and $1/2^j$. Each complete block contributes between $1/2$ and 1.

Complete the last partial block for an upper bound. There are $\lfloor \log_2 n \rfloor + 1$ blocks, each contributing at most 1. The first $\lfloor \log_2 n \rfloor$ complete blocks give a lower bound. For $n \geq 2$,

$$\frac{\lfloor \log_2 n \rfloor}{2} \leq H_n \leq \lfloor \log_2 n \rfloor + 1.$$

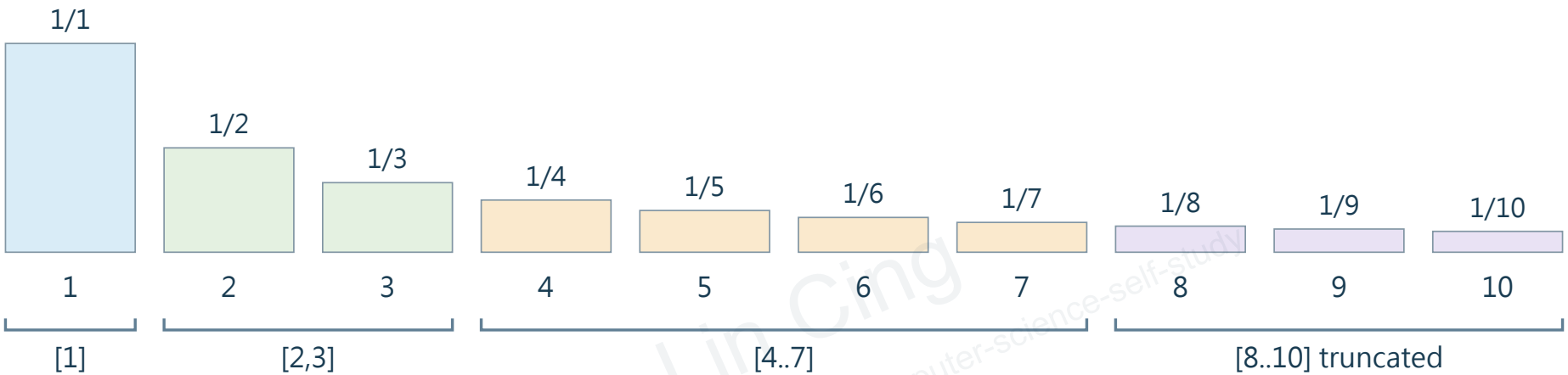
The lower bound is $\Omega(\log n)$, so $H_n = \Theta(\log n)$, also proving that the infinite series diverges.

At $n = 10$, the blocks $[1]$, $[2, 3]$, $[4, 5, 6, 7]$, and $[8, 9, 10]$ contribute 1, $5/6$, $319/420$, and $121/360$. Their total is $H_{10} = 7381/2520 \approx 2.928968$. The last block has three terms, not eight.

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Dyadic blocks reveal harmonic layers (continued)

n=10: each full dyadic block sums between1/2 and1



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Integral bounds follow from unit intervals

If f is integrable and nondecreasing on $[m-1, n+1]$, each integer k satisfies

$$\int_{k-1}^{k} f(x)\,dx \le f(k) \le \int_{k}^{k+1} f(x)\,dx.$$

Summing gives

$$\int_{m-1}^{n} f(x)\,dx \le \sum_{k=m}^{n} f(k) \le \int_{m}^{n+1} f(x)\,dx.$$

For a nonincreasing f, the inequalities reverse:

$$\int_{m}^{n+1} f(x)\,dx \le \sum_{k=m}^{n} f(k) \le \int_{m-1}^{n} f(x)\,dx.$$

The endpoint shifts control the error. A sum is not simply replaced by an integral. Since $1/x$ decreases for $x > 0$,

$$H_n \ge \int_{1}^{n+1} \frac{dx}{x} = \ln(n+1).$$

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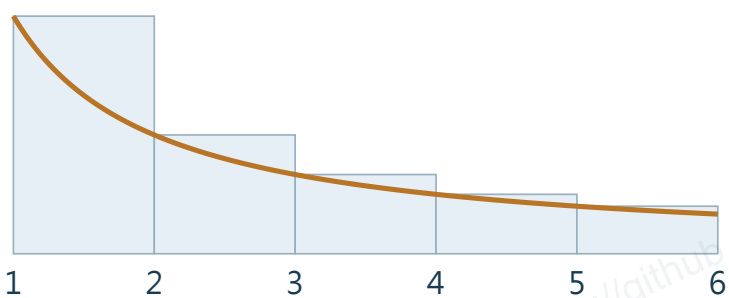
Integral bounds follow from unit intervals (continued)

Integrating from zero for an upper bound encounters a singularity. Remove the first term, then bound the terms $k = 2, \ldots, n$:

$$H_n \leq 1 + \int_1^n \frac{dx}{x} = 1 + \ln n.$$

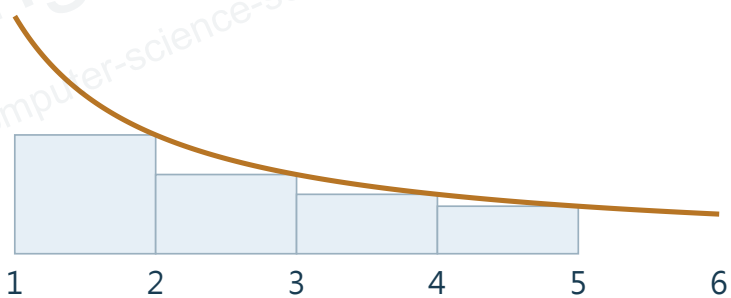
Decreasing1/x: height1/k, different endpoint intervals

Lower: $\int_1^6 1/x\,dx \leq H_5$



On[k,k+1]: curve≤1/k

Upper: $H_5 - 1 \leq \int_1^5 1/x\,dx$



On[k−1,k]: curve≥1/k

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Recomputable two-sided integral bounds

At $n = 5$, $H_5 = 137/60 \approx 2.283333$ lies between $\ln 6 \approx 1.791759$ and $1 + \ln 5 \approx 2.609438$. This bracket has an explicit width; it does not assert equality with $\ln n$. In general,

$$\ln(n+1) \le H_n \le \ln n + 1, \qquad H_n = \ln n + O(1).$$

For a fixed $p > 1$,

$$\sum_{k=1}^{n} \frac{1}{k^p} \le 1 + \int_1^n x^{-p}\,dx \le 1 + \frac{1}{p-1}.$$

The nonnegative partial sums increase and are bounded, so they converge. The case $p = 1$ diverges. For $0 \le p < 1$, the integral lower bound grows as n^{1-p}, and the sum is $\Theta(n^{1-p})$. Vanishing terms alone do not imply convergence.

For a fixed $r \ge 0$, integration also gives

$$\frac{n^{r+1}}{r+1} \le \sum_{k=1}^{n} k^r \le \frac{(n+1)^{r+1} - 1}{r+1}.$$

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Recomputable two-sided integral bounds (continued)

This again proves $\Theta(n^{r+1})$, with constants allowed to depend on the fixed parameter r.

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Nested loops and changing summation order

Loops with $i = 1, \dots, n$ and $j = i, \dots, n$ perform constant work

$$\sum_{i=1}^{n} (n - i + 1) = \frac{n(n+1)}{2}$$

times. Counting by j instead gives $i = 1, \dots, j$, hence $\sum_{j=1}^{n} j$. Both count the index set $\{(i, j) : 1 \leq i \leq j \leq n\}$.

If an iteration costs j, then

$$\sum_{i=1}^{n} \sum_{j=i}^{n} j = \sum_{j=1}^{n} \sum_{i=1}^{j} j = \sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}.$$

At $n = 3$, the row sums are $6, 5, 3$, totaling 14. The column count $1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3$ also totals 14.

Finite reordering is safe. An infinite double sum with signed terms requires absolute convergence before arbitrary interchange of order. Finite identities alone do not justify that operation.

Sources

Bibliography

Cormen, Leiserson, Rivest, Stein. *Introduction to Algorithms*, 3rd ed., MIT Press, 2009, Appendix A.

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References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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